

## Introduction

The intense luminosities of classical novae ( $\gtrsim 10^4 L_\odot$ ) and the short orbital periods ( $\sim$  hours) of the cataclysmic variable systems that host them indicate that significant irradiation of the donor star during a nova eruption and the post-outburst supersoft source (SSS) phase is unavoidable. Irradiation can lead to enhanced mass transfer, leading to a "bootstrapped" long-lived SSS phase that may explain some short-period SSS objects and have implications for the long-term evolution of the system.

## Methods

We model the combined cataclysmic variable system in MESA binary, which allows for evolving both stars simultaneously while tracking orbital parameters. We base the project off of the `evolve_both_stars` test case, to model a white dwarf (WD) and a main sequence (MS) star. For the WD, we use a modified version of the `inlist` from the `wd_nova_burst` test case for the WD and a fairly simple `inlist` for the MS star. For initial models, we use the end state of the `wd_nova_burst` test case, which has a mass of  $1.115 M_\odot$  and a zero-age main sequence companion of mass  $0.5 M_\odot$  using MESA's interpolation between a set of precomputed ZAMS models.

We set the initial orbital period to 6 hours (large enough to be initially detached) and employ the standard magnetic braking prescription available in MESA binary assuming conservative mass transfer. To handle irradiation of the MS star from the WD, we employ two controls in the binary `inlist`: `use_accretor_luminosity_for_irrad = .true.` (instructs MESA to use the actual luminosity from the accretor rather than an inferred accretion luminosity) and `col_depth_for_eps_extra = 1.5d0` (sets the depth over which irradiation is applied to the donor).

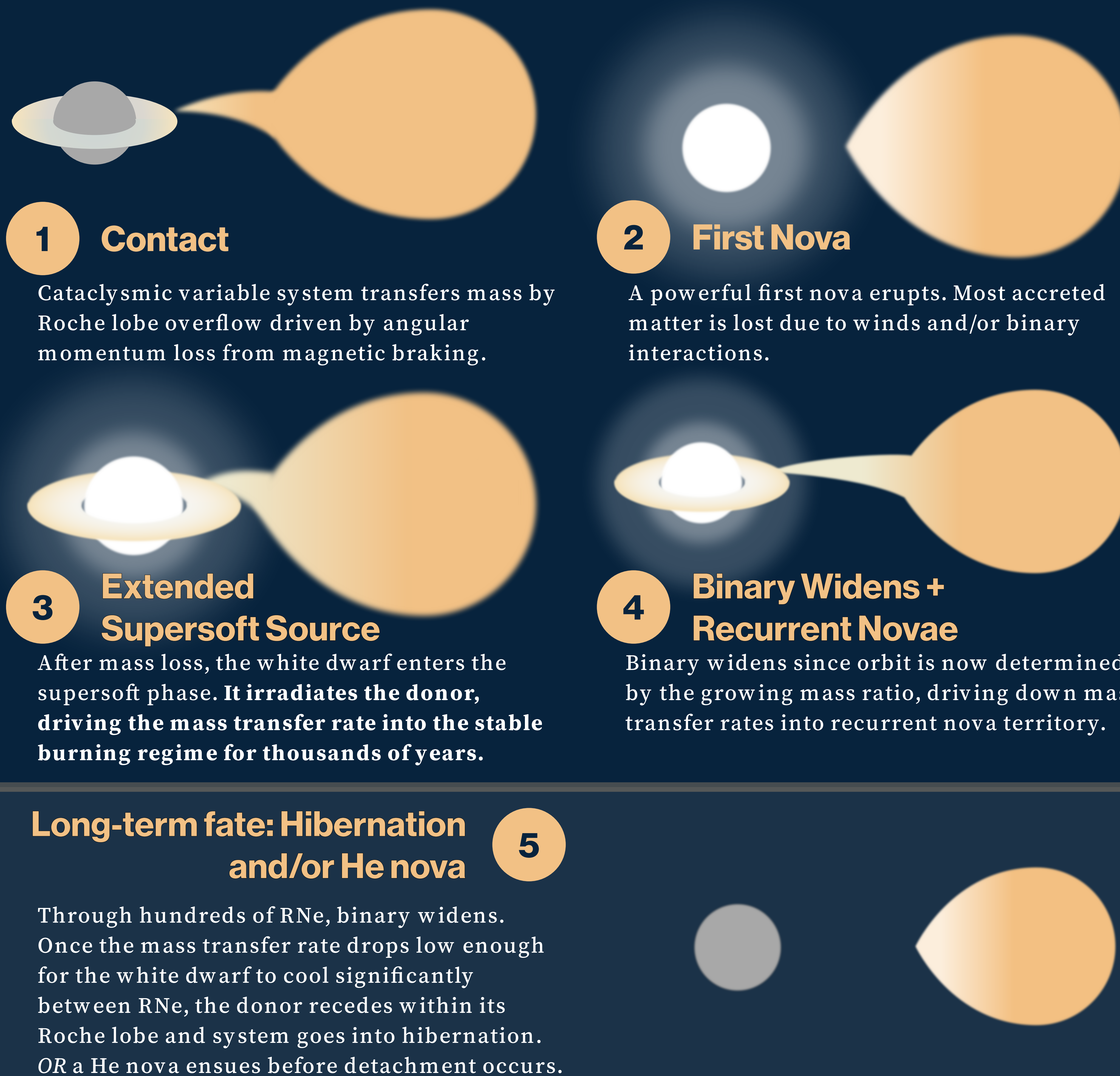
We evolve the system until there is strong helium burning. We're also exploring other combinations of masses and initial orbital periods, but this poster focuses on the parameters described above.

## Results

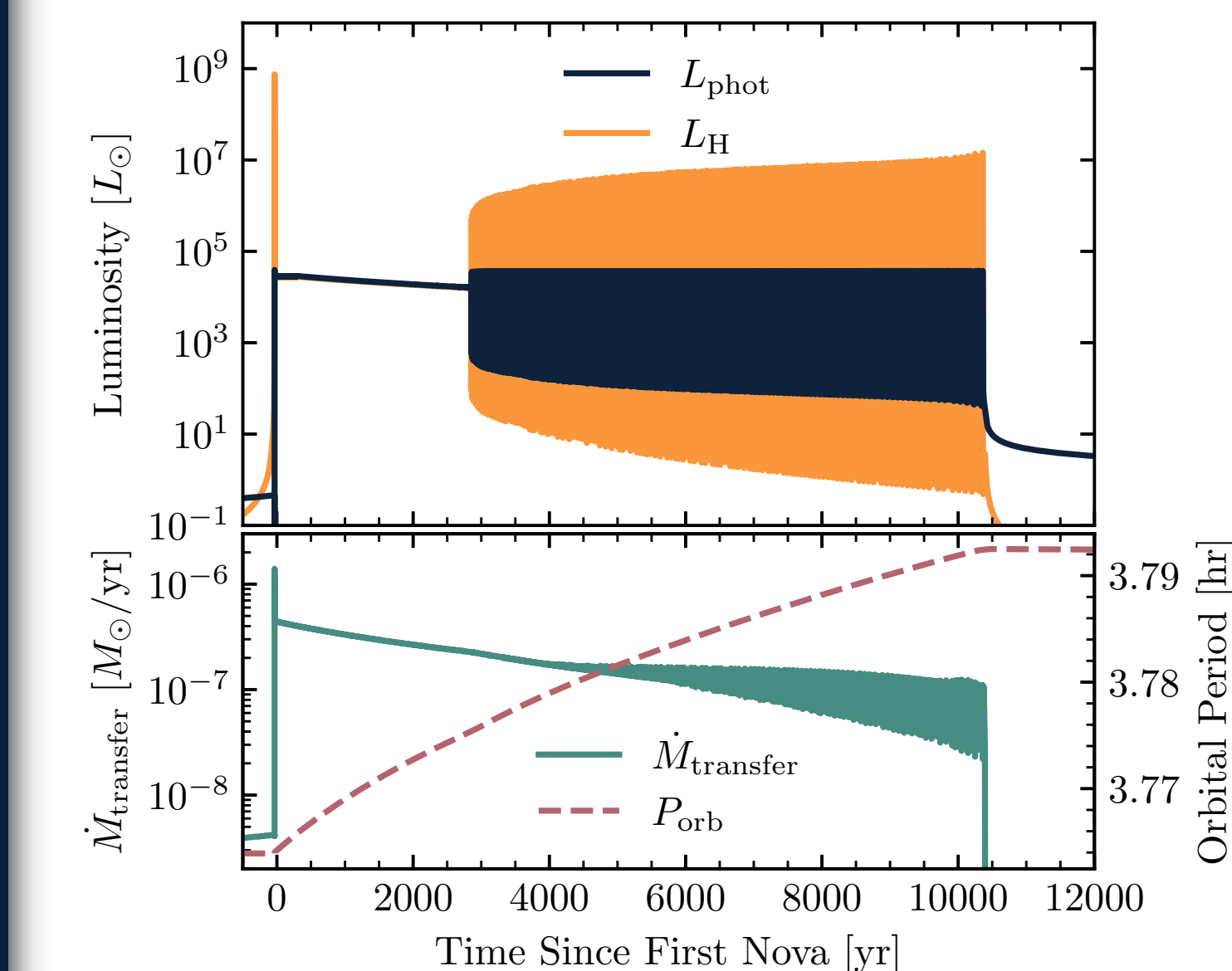
The binary starts detached, but at a period of ???, the MS star overflows its Roche Lobe, driving a mass transfer rate of  $\approx ???$ . After roughly ??? of hydrogen rich material is transferred, the WD undergoes a nova, but after the eruption subsides, the irradiation of the MS star is strong enough to drive mass transfer rates on the order of ???, causing the SSS phase to be extended far beyond the length of  $\approx ???$  expected without irradiation.

The SSS phase continues for ??? years, but since the transfer acts to make the mass ratio more extreme, the binary widens during this phase. This causes the Roche lobe of the MS star to increase while weakening the irradiation. Eventually, the mass transfer rate drops below the stability limit, and recurrent novae ensue, since the irradiation is still actively enhancing the mass transfer rate. For this model, after several hundred recurrent novae, the helium layer grows large and hot enough to trigger a large helium flash, at which point we terminate the evolution.

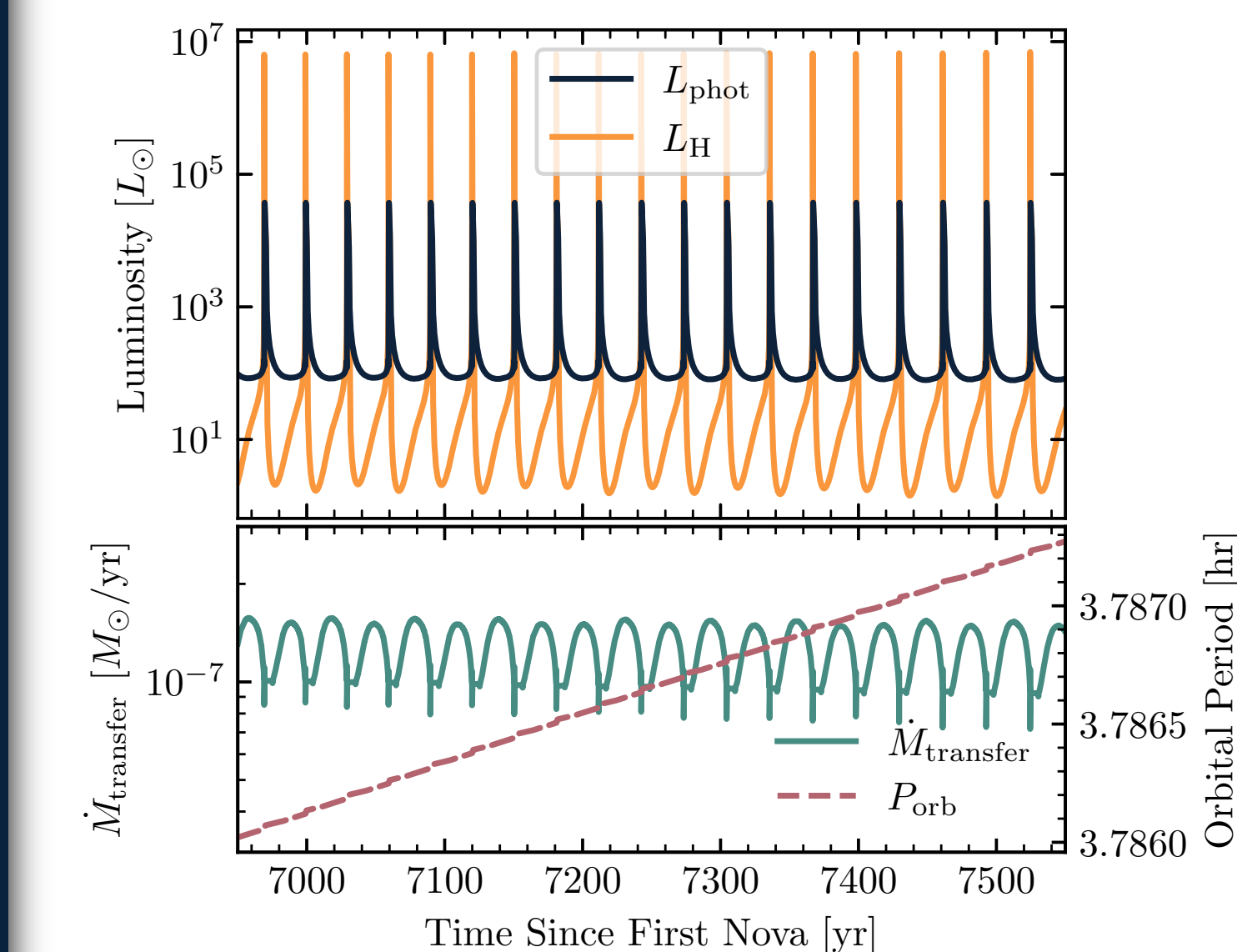
# Post-nova supersoft sources can extend their lifetimes by irradiating their donor stars.



## Figures



**Figure 1:** Evolution of a system comprising a  $1.115 M_\odot$  WD and a  $0.50 M_\odot$  MS companion during and after the first nova. *Top panel:* The photospheric luminosity and total power generated from hydrogen fusion. The initial spike is the first nova, which is followed by the SSS phase for nearly 3,000 years (simpler models without irradiation last less than 3 years). After that, the system enters a period of recurrent novae lasting around 7,000 years, followed by detachment and hibernation. *Bottom panel:* Orbital parameters for the same system. Note that the mass transfer rate is in the vicinity of  $10^{-7} M_\odot/\text{yr}$  through much of this phase, which make the extended SSS phase and RNe possible. The enhanced mass transfer from light to heavy object causes the orbital period to increase and the mass transfer rate to decrease, ultimately leading to hibernation.



**Figure 2:** Same system as Figure 1, but now zooming in on the RNe phase showing individual novae and the accompanying variation in  $\dot{M}$ .

